

# Diffusion from Random Motion: Exercise Guide for 1B

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Mathematical Methods for Physicists II

Python Notebook Guide

This is a collection of notes based on the random-walk and diffusion topics from my Mathematical Methods for Physicists II coursework. It accompanies 1B, whose exercises also include internship warm-ups and numerical extensions. It explains particle simulation, analytic diffusion, grid evolution and drift, preserving that distinction in the exercises' origins. The shared foundations guide should be read first.

## 1 Exercise 1: Release particles and record snapshots

### 1.1 Task and mathematical target

Release 5000 particles at zero, evolve for 400 independent fair steps, and save selected snapshots and widths. With length  $a$  and time per step  $\tau$ ,

$$D = \frac{a^2}{2\tau}, \quad \sigma_N = a\sqrt{N}.$$

The position array is the current ensemble, not one trajectory.

### 1.2 Develop the algorithm

1. Create one zero per particle, using floating-point storage for noninteger lengths.
2. Select snapshot step counts (defaults: 20, 100, 400).
3. At every step, move each particle independently left or right.
4. After all particles have moved, measure the ensemble standard deviation.
5. At selected counts, save a copy of the position array.

Begin the width history with zero; it then has  $N + 1$  entries. Loop indices count steps; physical elapsed time is the count multiplied by  $\tau$ .

### 1.3 Copies and checks

Snapshots need copies: storing a reference to the changing array would not freeze its values. At step  $n$ , sites obey parity and  $|X_n| \leq na$ . Saved snapshots should remain unchanged during later evolution. Sample mean fluctuates near zero because the ensemble is finite.

## 2 Exercise 2: Snapshot densities

### 2.1 Task and binning logic

Represent each particle cloud as a normalized density histogram. At fixed age, reachable sites are separated by  $2a$ : even counts reach  $2ja$  and odd counts reach  $a + 2ja$ .

1. Identify the even/odd offset.
2. Express positions as integer labels on that accessible lattice.
3. Find the minimum and maximum labels.
4. Place bin edges one step length either side of each site.

This produces width- $2a$  bins centred on reachable positions. Round lattice labels before converting to integers when floating-point arithmetic is used.

### 2.2 Normalization and interpretation

For bin count  $C_i$  and  $M$  particles,

$$h_i = \frac{C_i}{M(2a)}.$$

Its bar area estimates bin probability. Density has inverse-length units, and total bar area is one if all particles are included. Use the same position range across times and label titles with physical elapsed time. The cloud widens while its centre stays near zero.

## 3 Exercise 3: Analytic diffusion prediction

### 3.1 Task and mathematical target

For a unit point release at the origin on an infinite line,

$$p(x, t) = \frac{e^{-x^2/(4Dt)}}{\sqrt{4\pi Dt}}, \quad t > 0, \quad \text{Var}(X) = 2Dt.$$

The delta initial condition represents all initial probability at one point.

### 3.2 Develop the algorithm

1. Retrieve a snapshot and its step count  $n$ .
2. Convert to time  $t = n\tau$ .
3. Choose finely spaced plotting positions.
4. Evaluate the Gaussian density there.
5. Overlay it on a density-normalized particle histogram.

The plotting grid need not coincide with particle sites. Avoid evaluating the analytic formula at  $t = 0$ .

### 3.3 Checks and interpretation

The curve is symmetric and has width  $\sqrt{2Dt}$ . Default widths at 20, 100 and 400 steps are approximately 4.472, 10 and 20. Large step counts improve the coarse-grained Gaussian approximation. More particles reduce sampling noise but do not remove lattice parity.

## 4 Exercise 4: Width and variance over time

### 4.1 Task and mathematical target

Compare measured widths with

$$\sigma(t) = \sqrt{2Dt}, \quad \sigma^2(t) = 2Dt.$$

### 4.2 Develop the algorithm

1. Construct counts from zero through  $N$  and multiply by  $\tau$ .
2. Calculate the predicted width at each physical time.
3. Plot measured and predicted widths.
4. Square the measured widths and compare with  $2Dt$ .

Convert lists to numerical arrays before elementwise squaring when required.

### 4.3 Checks and interpretation

Both curves start at zero. Width grows as  $\sqrt{t}$ , variance linearly. Comparing the entire time dependence is stronger than checking one final value. Width squared is variance about the sample mean, not raw mean squared position:

$$\frac{1}{M} \sum_m X_m^2 = s_M^2 + \bar{X}^2.$$

Their difference is small for the fair ensemble, but matters under drift.

## 5 Exercise 5: Numerical diffusion on a grid

## 5.1 Task and numerical model

Evolve the density itself rather than individual particles. Choose spacing  $\Delta x$ , numerical time step  $\Delta t$ , and a finite domain. Numerical time steps need not equal the particle interval  $\tau$ , but both calculations must end at the same physical time. The domain extends to at least five predicted final standard deviations on either side. Choose an integer number of updates and adjust  $\Delta t$  to reach the target exactly.

## 5.2 Initial probability

Initialize densities to zero and set the central value to  $1/\Delta x$ , giving

$$\sum_i p_i \Delta x = 1.$$

A central value of one gives unit mass only when  $\Delta x = 1$ .

## 5.3 Derive and implement the update

Forward time and centred spatial differences give

$$p_i^{n+1} = p_i^n + r(p_{i+1}^n - 2p_i^n + p_{i-1}^n), \quad r = \frac{D\Delta t}{(\Delta x)^2}.$$

Require  $0 \leq r \leq 1/2$ . The defaults  $D = 0.5$ ,  $\Delta x = 1$ ,  $\Delta t = 0.4$  give  $r = 0.2$ . For every update:

1. Copy the old density array.
2. Loop over interior sites and evaluate the formula from old values.
3. Leave endpoints at zero.
4. Replace the old array only after every interior value has been calculated.

Updating in place mixes old and new neighbors and changes the method.

## 5.4 Boundaries, checks and comparison

Fixed-zero endpoints are absorbing boundaries. The analytic Gaussian assumes an infinite line. Far-away edges make the central comparison close, but allow a small probability loss. Check  $p_i \geq 0$  and remaining probability  $\sum_i p_i \Delta x$ . Compare the final grid density, particle histogram and analytic curve at the same physical time. Agreement is expected in macroscopic shape and width, not necessarily lattice-scale structure. A stronger accuracy check refines the grid while preserving stability and measures convergence to the analytic solution.

# 6 Follow-up: A biased walk

## 6.1 Task and new concept

Give a particle right-step probability  $q = 0.65$ . Drift means motion of the cloud's centre; spreading is measured about that centre.

## 6.2 Mathematical target

The mean increment is  $a(2q - 1)$  and its variance is  $4a^2q(1 - q)$ . Thus,

$$E[X_N] = Na(2q - 1), \quad \sigma_N = a\sqrt{4Nq(1 - q)}.$$

With default units at 400 steps, the mean is 120 and width  $\sqrt{364} \approx 19.08$ .

### 6.3 Develop the algorithm

1. Initialize a separate particle ensemble at zero.
2. Draw a uniform number in  $[0, 1)$  for each particle and step.
3. Move right if it is below  $q$ ; otherwise move left.
4. Record the mean and width after all particles have moved.
5. Calculate theoretical curves in physical time.
6. Compare fair and biased endpoints using one shared set of histogram bins.

Shared bins span both endpoint arrays and make the density comparison direct.

### 6.4 Interpretation

Define

$$v = \frac{a(2q - 1)}{\tau}, \quad D_{\text{spread}} = \frac{2a^2q(1 - q)}{\tau}.$$

Then  $E[X] = vt$  and  $\text{Var}(X) = 2D_{\text{spread}}t$ . The continuous large-step-count approximation needs both terms:

$$p_t = -vp_x + D_{\text{spread}}p_{xx}.$$

Its spreading coefficient differs from the fair value. Raw mean squared displacement includes centre motion:

$$E[X^2] = 2D_{\text{spread}}t + (vt)^2.$$

Use the mean to measure movement of the centre and variance to measure cloud broadening. Finite samples fluctuate around both predictions.